

Shaun Sekura

EDUCATION

The Ohio State University

Columbus, OH

Bachelor of Science in Mechanical Engineering

Expected December 2027

GPA: 3.404/4.0 — Recent Coursework: Machine Elements, Kinematics, System/Vibrational Dynamics, Circuits, Thermodynamics, Machine Design, Numerical Methods, Mechanics of Materials, Linear Algebra

TECHNICAL SKILLS

- **Software:** SolidWorks, CATIA, Onshape, Python, MATLAB, Excel, Arduino IDE, EES
- **Lab Research/Processes:** CAD Software, FEA Analysis, Machining, Equipment Operation
- **Fabrication:** CNC Machining, Vertical Mill, Lathe, 3D Printing, Soldering/Wiring (Arduino), Drafting/GD&T

EXPERIENCE

Undergraduate Teaching Assistant

The Ohio State University

Intro To Mechanical Design (MECHENG 2900)

Spring 2026 – Present

- Mentored students in circuit construction and embedded systems programming.
- Communicated complex engineering concepts clearly to students with varying technical backgrounds.

EcoCAR Competition Team

Center For Automotive Research

Volunteer Undergraduate Researcher

Fall 2025 – Present

- Designed and analyzed 10+ vehicle components using SOLIDWORKS simulations to test and improve strength-to-weight ratios and general vehicle design.
- Collaborated across sub-teams (electronics, efficiency, automation, etc.) to integrate structural designs into final build for testing.

Formula SAE

Center For Automotive Research

Volunteer Undergraduate Researcher

Fall 2024 – Spring 2025

- Assisted with mechanical design, fabrication, and assembly of a Formula SAE race car.
- Participated in vehicle testing and iterative improvements to support performance and reliability.

American Society of Mechanical Engineers (ASME)

The Ohio State University

Executive Board

Fall 2024 – Present

- Organized social events and design challenges, increasing club participation by 320% compared to previous years.
- Coordinated industry speaker series and professional networking events for 100+ members.

Humanitarian Engineering Scholars

The Ohio State University

Leadership Council

Fall 2024 – Present

- Spearheaded program engagement efforts by doubling the number of events hosted and tripling student attendance compared to the previous year.

Valleaire Golf Club

Hinckley, OH

Maintenance

2024 – Present

- Operated heavy-duty machinery to maintain a golf course spanning over 150 acres.
- Reduced water usage on the golf course by 15% saving the company \$4,000 through sustainable irrigation practices.

PROJECTS

Sustainable Rocket Redesign

- Used Python (AeroSandbox library) to build a reduced order simulation of the rocket's descent.
- Used SOLIDWORKS to reverse engineer a simplified model of the top of Starship.

Reverse Engineering of Motor Mount

- Reverse engineered a Cadillac Lyriq motor mount and worked with team to reinstall the part into the car.
- Used CATIA V5 to design mount and performed FEA Analysis to determine stress on mount.

INTERESTS

- Hiking, Music, Tennis, Basketball, Strategy Games, Competitive Trivia, Sushi